Publication-Level Generalization in Perovskite Solar Cell Prediction: DOI-Grouped Evaluation with Literature-Derived Recipe Features

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Abstract

Machine-learning (ML) models for perovskite solar cell (PSC) performance are increasingly trained on literature-derived databases such as the Perovskite Database Project (PDP), yet the PDP studies that describe their partitioning use record-level splits, even though a single publication often contributes many closely related device records. We build a power conversion efficiency (PCE) prediction framework on 41,107 PDP device records (7,326 DOI-identified publications) and evaluate it on unseen publications. Three nested feature tiers are compared: PDP-derived device, processing and composition features ($\mathcal{M}_1$); $\mathcal{M}_1$ plus curated material-property descriptors ($\mathcal{M}_2$); and $\mathcal{M}_2$ plus 36 LLM generated binary flags ($\mathcal{M}_3$) that are built deterministically from cached outputs of a locally served large language model (LLM) applied to full-text papers and are available for 36.6% of records. Three findings emerge. First, record-level random splitting gives a CatBoost R^2 of 0.59–0.62, against 0.37–0.41 under DOI-grouped splitting of the same data, features and model family, whereas a ridge model shows a gap of only 0.06; 69% of PCE variance lies between publications. DOI-grouped evaluation is therefore the more conservative estimate of performance on new publications. Second, material-property augmentation raises held-out performance (R^2 0.371 to 0.403 on 14,125 reserved records from 2,619 publications), but a material-identity control reaches the same R^2 (0.406 versus 0.407), so most of the measured gain reflects knowing which materials are present. Third, adding the recipe flags improved all three point metrics on the same reserved publications under matched settings (R^2 0.403 to 0.406, MAE 2.955 to 2.949 and RMSE 3.794 to 3.787 percentage points on all records), with a larger improvement on the 5,460 records whose publications have extraction data (R^2 0.359 to 0.366; $\Delta R^2 = +0.0071$, 95% publication-cluster bootstrap interval -0.0007 to $+0.0157$). The improvement is small, its interval includes zero, and it was measured on one reserved partition, so its magnitude and generalizability remain uncertain. Forward-in-time evaluation lowers R^2 to 0.345, and no external validation was possible.

Keywords: Perovskite solar cells; Power conversion efficiency; Materials informatics; Large language models; Publication-grouped validation; Data leakage.

1. Introduction

Perovskite solar cells (PSCs) are promising photovoltaic devices whose performance depends on the absorber material, device architecture, and fabrication conditions. A typical planar PSC consists of five functional layers: a substrate, an electron transport layer (ETL), a perovskite absorber, a hole transport layer (HTL), and a metal electrode. As illustrated in Fig. 1, two widely used configurations are n-i-p and p-i-n, which differ in the relative positions of the ETL and HTL[1,2]. Under illumination, the perovskite absorber generates charge carriers, while the ETL and HTL selectively transport electrons and holes, respectively, towards the electrodes. Charge extraction and recombination, together with the properties of the constituent materials, influence the resulting photovoltaic performance[1,2].

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Power conversion efficiency (PCE) is the principal metric of PSC performance and represents the fraction of incident optical power converted into electrical power. The fill factor (FF) characterizes the shape of the current–voltage curve as the ratio of the maximum output power to the product of the open-circuit voltage and short-circuit current. These quantities are expressed as

$$\mathrm{PCE} = \frac{V_{\mathrm{OC}} J_{\mathrm{SC}} \mathrm{FF}}{P_{\mathrm{in}}} \times 100\%, \tag{1}$$

and

$$\mathrm{FF} = \frac{V_{\mathrm{MPP}} J_{\mathrm{MPP}}}{V_{\mathrm{OC}} J_{\mathrm{SC}}}, \tag{2}$$

where V_{OC} and J_{SC} denote the open-circuit voltage and short-circuit current density, respectively; V_{MPP} and J_{MPP} are the voltage and current density at the maximum power point; and P_{in} is the incident optical power density[2]. The dependence of PCE on multiple interacting material, architectural, and processing variables makes systematic device optimization challenging. Machine learning (ML) offers a data-driven approach to identifying these relationships and exploring combinations of experimentally reported parameters that would be difficult to investigate exhaustively through conventional trial-and-error approaches[3–7].

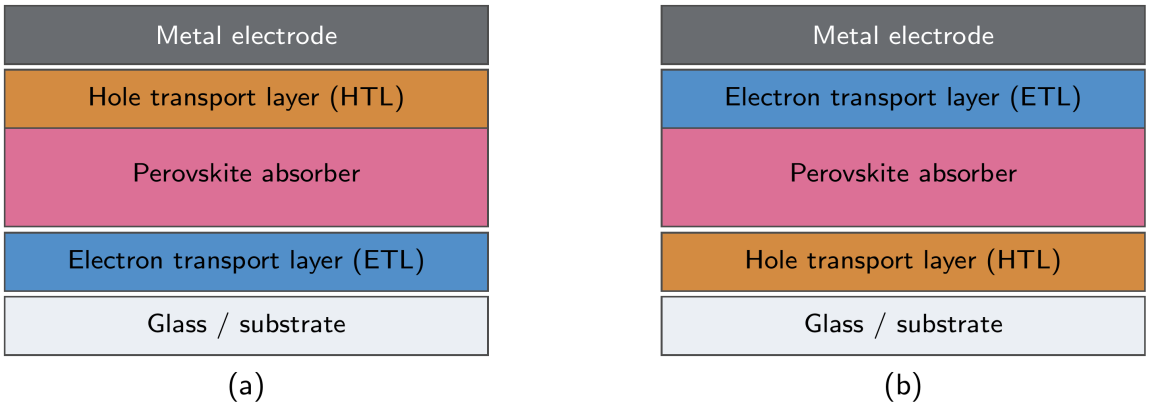


Figure 1: Example device structure for a (a) n-i-p and (b) p-i-n perovskite solar cell.

The availability of large experimental databases has enabled ML models to be developed using combinations of material, device, and process information. The Perovskite Database Project (PDP) provides an open-access collection of PSC data covering material compositions, device structures, fabrication conditions, and photovoltaic characteristics[8]. Roberts et al. developed an open-source ML pipeline using the PDP and evaluated XGBoost, artificial neural networks, and Gaussian process regression for predicting short-circuit current density. Their XGBoost model achieved a test R^2 of 0.35 and an RMSE of 3.58 mA cm^{-2}, while their analysis highlighted the importance of data cleaning, feature encoding, and reproducible workflows[9]. Other device-level studies have similarly examined the influence of descriptor selection and dataset characteristics on predictive performance[10].

Fukasawa et al. examined the representation of material and process information specifically for PCE prediction using the PDP[11]. Starting with 42,459 records, they curated the dataset to obtain 36,937 records with 248 material- and process-related variables. Their results demonstrated that incorporating process information improved prediction compared with using perovskite composition alone. Under an 8:2 train–test split (the article does not describe grouping by publication), their best feature representation achieved a test R^2 of 0.7177 under the conventional split; evaluation on 294 newly registered records yielded an R^2 of 0.45 and an MAE of 3.20 percentage points[11]. They also identified substantial data degeneracy, with 6,446 unique degenerate points corresponding to 20,695 feature vectors, indicating that distinct experimental records can have identical feature representations despite exhibiting different PCE values.

The distinction between interpolation within an existing dataset and generalization to new groups of observations is also recognized in the broader materials-informatics and machine-learning literature. Meredig et al. introduced leave-one-cluster-out cross-validation to examine how evaluation strategies affect estimates of model performance for materials discovery, emphasizing that conventional random validation may not adequately represent extrapolative use cases[12]. More broadly, Kapoor and Narayanan documented data leakage as a recurring source of overoptimistic findings and reproducibility problems across scientific machine-learning applications[13]. These studies establish that evaluation should reflect the intended generalization setting and the dependence structure of the data. In literature-compiled PSC datasets, publication DOI provides a practical grouping variable for assessing performance on papers excluded from training. Our work applies this principle to PCE prediction and quantifies how the estimated performance and feature-tier comparisons change when publication-level overlap is removed.

The literature-derived nature of PSC databases raises a further concern about model evaluation. Multiple devices reported in the same publication may share material selections, fabrication procedures, device architectures and measurement conditions, so record-level independence is an assumption that requires scrutiny. Of the PDP-based studies we examined, the two that describe their partitioning use record-level splits[9,11], and a further PCE study does not state its splitting procedure[14]. The fall of Fukasawa et al.'s R^2 from 0.7177 to 0.45 on newly registered records is prior evidence that performance estimates are sensitive to the evaluation set, but that evaluation was small and was not a systematic DOI-grouped protocol. Group-aware, cluster-based, forward and time-split validation are established in materials informatics, cheminformatics and ecology[12,15–19]; the contribution here is narrower. To our knowledge, no earlier study compares record-level and DOI-grouped evaluation on the same PDP export under matched features, preprocessing and model family (our literature search was not exhaustive), and we do so here. DOI grouping prevents records of one publication from appearing in both training and test partitions and so directly assesses generalization to publications excluded from training.

A second line of work concerns the information held in the full text of the papers themselves. Large language models (LLMs) have been used to extract structured materials data from scientific text: Dagdelen et al. fine-tuned pretrained LLMs for joint entity and relation extraction on three materials-chemistry tasks[20]; Polak and Morgan proposed a prompt-engineering workflow with conversational LLMs and follow-up questions that reached precision and recall close to 90% on materials-data tests[21]; and Polak et al. described a model-agnostic extraction method built on general-purpose language models[22]. These studies contribute and evaluate extraction methods. Our use of a locally served LLM is application-specific and makes no claim about extraction quality relative to them. We ask a different question, namely whether structured recipe flags derived from such extractions improve the prediction of device PCE when whole publications are held out.

In this work we build a PCE prediction framework on the PDP and evaluate it at the publication level. The contributions are three.

(i) **Publication-level evaluation.** We compare record-level random splitting with repeated DOI-grouped holdouts under matched predictor definitions, model family and preprocessing, with DOI normalisation, an explicit treatment of records lacking a DOI, and a DOI-disjoint validation set so that no model selection touches the test publications. We quantify the gap for a flexible (CatBoost) and a linear (ridge) model, relate it to the within-publication structure of the data, and add lead-author and forward-in-time protocols.

(ii) **Material-property augmentation and a material-identity control.** Curated material-property descriptors ($\mathcal{M}_2$) improve held-out predictions over the PDP-derived baseline ($\mathcal{M}_1$). A control that supplies only the identities of the transport and absorber materials reproduces essentially the whole gain, which separates the empirical benefit from the stronger claim that property values carry independent transferable information.

(iii) **LLM-derived recipe flags.** Thirty-six binary recipe flags derived deterministically from cached LLM extractions ($\mathcal{M}_3$) give positive observed changes in R^2, MAE and RMSE over $\mathcal{M}_2$ on identical reserved publications under matched settings, with larger gains on the publications for which extractions exist.

The intervals include zero, so the magnitude and generalizability of the effect remain uncertain, and coverage is limited to 36.6% of records.

SHAP-based interpretation of an earlier model variant, and the results for the earlier literature representation, are given in the Supplementary Information. The overall workflow is summarized in Fig. 2.

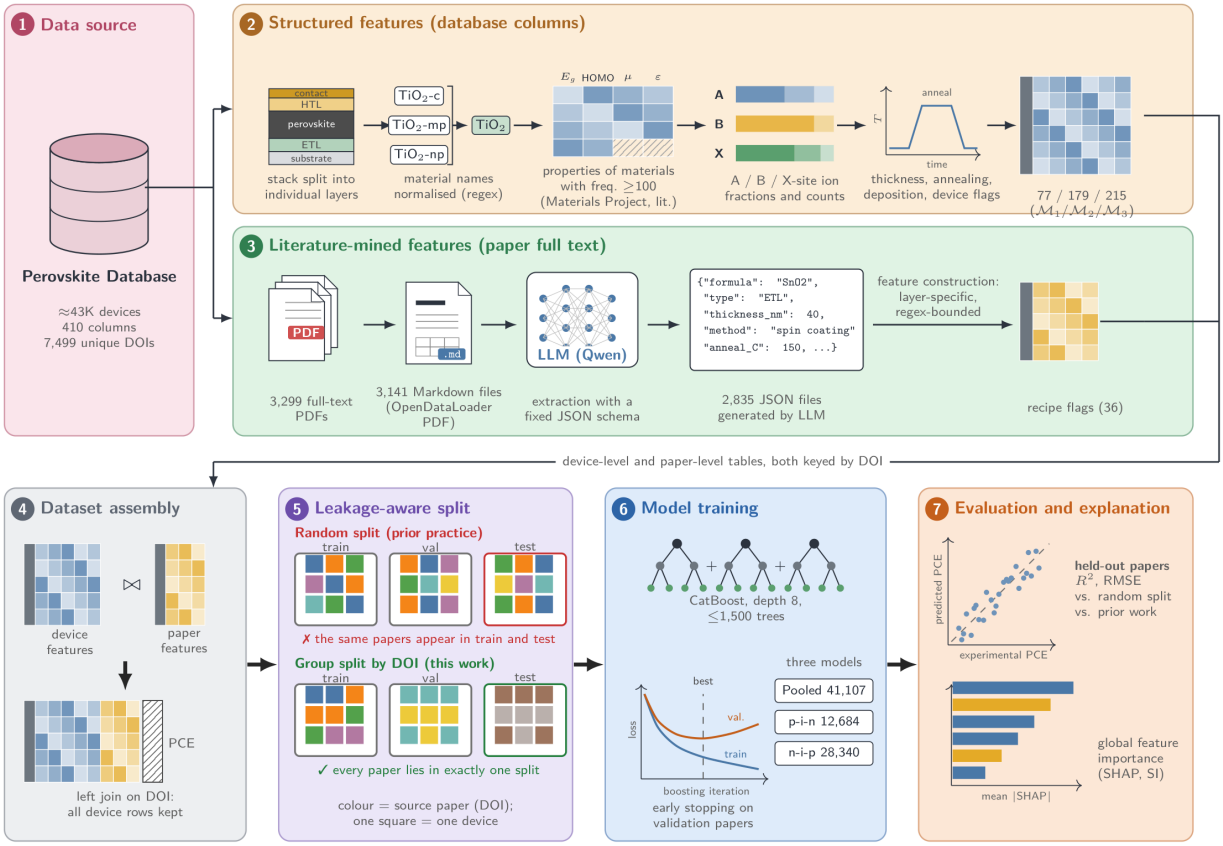


Figure 2: Overview of the machine-learning workflow for perovskite solar-cell power conversion efficiency (PCE) prediction. The Perovskite Database Project provided unique DOIs, from which full-text PDFs were retrieved, which were converted to Markdown. Subsequently structured JSON files were generated through large language model (LLM)-based extraction. The 36 recipe flags built from these files were integrated with structured database descriptors at the DOI level, followed by publication-grouped model evaluation using CatBoost. The predictor counts in panel 2 are for $\mathcal{M}_1$, $\mathcal{M}_2$ and $\mathcal{M}_3$. Panel 6 shows the matched CatBoost configuration and the subset sizes used.

2. Methods

2.1. Data Preparation

2.1.1. Dataset Description

The Perovskite Database Project (PDP) is an open-access repository of experimentally reported perovskite solar cell (PSC) data, encompassing material compositions, device architectures, fabrication procedures, and photovoltaic characteristics[8]. The database export used in this study contained 43,398 device records, associated with 7,499 unique DOIs, and 410 columns. These records comprise heterogeneous data types, including categorical descriptions of functional layers, perovskite compositions, numerical device parameters, and fabrication-related variables. Previous PDP-based studies have demonstrated that appropriate data selection, cleaning, and representation are important for developing predictive models from these heterogeneous records[9,11].

Five fields were initially extracted to construct the layer-resolved material representation: the ETL and HTL stack sequences, the perovskite short-form composition, the PCE field `JV_default_PCE`, and the cell architecture. The target is the PDP field `JV_default_PCE`, renamed `final_pce`, which the PDP documents as the "selected power conversion efficiency" of the device in its current–voltage section; the PDP describes the corresponding default V_{OC} as the best or stabilised value. The export carries no further documentation of how the default value is chosen among forward, reverse or stabilised measurements, and we make no assumption about it beyond this definition: the target is the PCE reported as the default value for each device record. The architecture field was retained for pooled modelling and architecture-specific analysis. This extract was an intermediate representation for material processing rather than the complete predictor matrix; additional device and fabrication variables were retained from the original database for feature construction. Publication metadata, including DOI identifiers and author information, were excluded from the predictors, while DOI identifiers were retained separately for linking records to the literature corpus and defining publication-level data partitions. A field-by-field audit of the predictor matrices of all three tiers (Supplementary Table S8) found no other PCE measurement, forward or reverse scan metric, stabilised-performance, EQE-derived, stability or other J–V field among the predictors; the five outcome-adjacent columns of the processed table (`Stability_measured`, `EQE_measured`, `JV_hysteresis_index`, `Cell_area_measured_cm2` and `Stabilised_performance_measured`) are excluded in code.

2.1.2. Layer-Stack Parsing and Material Catalogue

The material-stack descriptions in the PDP are stored as strings containing multiple materials or sequential layers, with vertical bars (|) delimiting successive entries. These strings were parsed into position-indexed fields, such as `ETL_1`, `HTL_1`, and `Perovskite_1`, making the constituent materials independently accessible. The resulting representation accommodated up to seven ETL entries, three HTL entries, and five perovskite-layer entries. Where an entry contained multiple semicolon-separated materials, the individual materials were expanded into separate fields while preserving the recorded stack order. Material identifiers were also normalized by consolidating selected naming variants into common identifiers, facilitating consistent matching across device records.

A catalogue of unique standardized material identifiers was subsequently constructed from the parsed ETL, HTL, and perovskite fields. The catalogue contained 2,817 unique material identifiers, and the occurrence frequency of each identifier was calculated by counting its appearances across the parsed stack fields. This catalogue provided a common reference for material-level property retrieval and the construction of layer-specific descriptors (Section 2.3.2). The parsed identifiers were used for property lookup and were not themselves supplied to the models as categorical predictors.

2.2. Literature Corpus Construction

2.2.1. DOI Identification and Full-Text Retrieval

To enrich the structured device database with information from the original experimental literature, publication identifiers were taken from the `Ref_DOI_number` field of the PDP. The export contains 7,499 distinct non-empty DOI strings, which correspond to 7,454 publications after DOI normalisation (Section 2.4), and 286 records without a DOI. Papers were identified by DOI, and full texts were retrieved through the institutional network using the subscriptions and open-access routes available to the authors' institution. Where automated retrieval failed, some papers were obtained manually through institutional access. The route by which each individual PDF was obtained was not logged, so we do not claim that it was verified per document. Retrieved documents were associated with their DOIs to keep traceability through the later processing steps. The retrieval stage yielded 3,299 PDF documents; because availability and retrieval success varied across publications, the corpus is the subset of identified publications for which a PDF was obtained. The PDFs, their Markdown conversions and the raw LLM output files are not redistributed (Data and code availability).

2.2.2. PDF Conversion and LLM-Based Information Extraction

The retrieved PDF documents were converted individually into Markdown using OpenDataLoader PDF, an open-source PDF parsing tool designed to produce machine-readable representations of document content

and structure[23]. This conversion yielded 3,141 Markdown documents from the 3,299 retrieved PDFs. The resulting files provided a text-based representation of each publication for subsequent processing by a large language model (LLM).

Each Markdown document was supplied individually to a locally served Qwen3.5-9B model (OpenAI-compatible vLLM endpoint, model alias `qwen35`, temperature 0.1, at most 16,384 output tokens; the exact checkpoint and quantisation are not recorded in the project files) using a predefined system prompt (reproduced in the Supplementary Information) and a structured JSON schema to guide consistent information extraction. The recovered extraction script is zero-shot; whether additional examples were used for the 2,835 outputs cannot be verified, so no few-shot claim is made. The extraction procedure was designed to identify reported materials and their functional classifications, together with available material properties, units, uncertainties, and measurement methods. Fabrication-related information, including deposition techniques, spin-coating parameters, annealing conditions, and processing environments, was also extracted where reported. Photovoltaic performance information was collected separately, with champion and average performance values distinguished where possible. The resulting structured outputs retained their association with the source DOI for subsequent paper-level feature construction.

The individual extraction outputs were consolidated into a combined JSON representation and subsequently organized into DOI-indexed JSON files. This process yielded 2,835 structured paper records. These records served as the intermediate representation for downstream feature construction, in which the extracted information was normalized and converted into a consistent set of publication-level descriptors for integration with the device-level database.

No hand-annotated reference set was available, so extraction precision, recall and hallucination rate were not measured, and the checkpoint and quantisation of the Qwen model are not recorded. As a proxy we compared an earlier table of extracted features with the independently curated PDP entries for the same publications. For the numerical parameters, the paper-level value agreed with the median PDP device value within 10% for 59–85% of publications that have both (Spearman 0.44–0.90; perovskite thickness 0.79, $n = 507$; annealing temperature 0.71, $n = 455$), and for material flags the DOI-level precision and recall against the PDP stack strings were 0.94/0.82 (TiO_2), 0.95/0.80 (Spiro-OMeTAD), 0.82/0.87 (PEDOT), 0.85/0.83 (NiO_x), 0.83/0.83 (PTAA) and 0.54/0.82 (SnO_2), with lower precision where the PDP records only a subset of the materials a paper describes. This is agreement with a second curated source, not accuracy against ground truth, and it was computed for the earlier feature table and not for the 36 recipe flags of $\mathcal{M}_3$, which have not been validated against a manually annotated gold set. The extracted numerical values equal the PDP value exactly in 59–82% of records where both exist and supply a value where the PDP column is empty for only 0.8–5.3% of records, so they are largely redundant with columns already present in $\mathcal{M}_1$. A blank annotation worksheet (80 stratified publications, with each flag definition, extracted value and evidence span) has been prepared for human review, but no review has been carried out and no validation accuracy is claimed.

The corpus shrank from 3,299 retrieved PDFs to 3,141 Markdown documents and 2,835 structured paper records. The two literature representations used in this work have slightly different coverage masks. The recipe flags of $\mathcal{M}_3$ are non-missing for 15,027 of the 41,107 filtered device records (36.6%), from 2,731 DOI-associated publications. The earlier literature representation covers 14,896 records (36.2%) from 2,699 publications. The 32-publication (131-record) difference arises because 32 extraction records contain no extracted materials: the earlier table is missing for them, whereas the recipe-flag table keeps a row with all 36 flags equal to zero. For those records the zero flags therefore reflect an empty extraction rather than the absence of the recipes, and the "covered" population in the recipe-flag evaluation means that an extraction record exists. Why these extractions are empty could not be determined. In both tables the features are non-missing exactly for publications with an extraction record (individual sparsely reported process parameters of the earlier table have lower coverage, 3.7–36.2%) and, where present, are constant within a publication.

2.2.3. Earlier Literature Representation (Supplementary Information)

Before the recipe flags, a broader 44-predictor literature representation (32 base columns plus missingness indicators; material and layer flags, thicknesses, annealing parameters and extraction-completeness counts)

was built from the same extraction JSON by recursive traversal of the nested records, with synonym normalisation and bounded pattern matching. It is publication-level, was superseded by the recipe flags, and is not part of $\mathcal{M}_1$, $\mathcal{M}_2$ or $\mathcal{M}_3$. Its ablation, controls and SHAP analysis are reported in the Supplementary Information as an earlier feature representation and are not used to draw conclusions about $\mathcal{M}_3$.

2.3. Feature Engineering

2.3.1. Composition-Derived Descriptors

Composition-derived descriptors were constructed to represent the elemental and ionic composition of the perovskite absorber numerically. These features included the fractions of constituent ions at the A, B, and X sites and corresponding ion counts.

The composition-derived descriptors of the $\mathcal{M}_2$ material-property block (Section 2.3.2) include the average atomic mass and average atomic radius of the perovskite absorber, which are read from the material catalogue and are not part of $\mathcal{M}_1$. The feature `Perovskite_avg_atomic_mass` was calculated as a stoichiometric average of the atomic masses of the constituent elements. For a composition containing k constituent elements, this descriptor is defined as

$$\overline{M} = \frac{\sum_{i=1}^{k} n_i M_i}{\sum_{i=1}^{k} n_i},$$

where n_i is the stoichiometric coefficient of element i in the composition and M_i is its atomic mass. Consequently, the descriptor is the average atomic mass weighted by the elemental stoichiometry of the catalogue formula rather than an unweighted average of the constituent atomic masses.

The perovskite geometry descriptors were computed from the ion-fraction columns with fixed effective radii (pm): A-site FA 253, MA 217, Cs 188, Rb 172; B-site Pb 119, Sn 110, Bi 103; X-site I 220, Br 196, Cl 181, F 133. Each site radius is the fraction-weighted mean over the species present on that site, normalised by the summed fractions of the listed species. For an idealised ABX_3 composition the octahedral factor is then

$$\mu = \frac{r_B}{r_X},$$

with r_B and r_X the fraction-weighted site radii; the tolerance factor $t = (r_A + r_X)/[\sqrt{2}(r_B + r_X)]$, the ratio r_A/r_X and a τ descriptor were computed from the same radii. We recomputed `perovskite_octahedral_factor` and `perovskite_tolerance_factor` from the stored fraction columns with these radii and reproduced the stored values to floating-point precision. The descriptors are defined only when all three sites have at least one listed species, and they ignore species outside the lists (for example organic cations other than FA and MA). Their inclusion does not by itself establish structural stability or a causal relationship with PCE.

By contrast, the layer-level `Perovskite_avg_radius` (and the corresponding ETL and HTL columns) is *not* an ionic radius. In the recovered script that fills the material catalogue it is the atom-count-weighted mean of covalent atomic radii (pm; for example C 77, H 31, N 71, Pb 146, I 139) over the elements of a hand-entered formula for the material, so that $MAPbI_3$ (CH_6NPbI_3) gives 74.75 pm. The same script computes the minimum, maximum and average atomic mass and the Pauling electronegativity. These columns are therefore stoichiometric composition summaries; they are available only for materials with an entered formula (the average radius is missing for many mixed perovskite compositions) and should not be read as ionic-size descriptors.

2.3.2. Material-Property Augmentation

A separate material-property reference catalogue was curated for the standardized material identifiers that occur at least 100 times across the parsed ETL, HTL and perovskite stacks; materials below this threshold have missing descriptors. Band gaps, HOMO and LUMO energies, electron and hole mobilities and dielectric constants were compiled from published literature and, where applicable, retrieved from the Materials Project database; the atomic-mass, atomic-radius and electronegativity summaries are computed from the stoichiometry of each material's formula as described above. Source references were stored with

the curated values in the catalogue. The catalogue file itself is not part of the supplied package, so the source of an individual band-gap or HOMO/LUMO value, and the split between literature and database values, cannot be audited from the supplied files. The values are general material-level reference properties and not measurements on the specific devices in which the materials appear, and property availability is heterogeneous.

For each device, property lookup was performed separately for the ETL, HTL and perovskite layers. Within each layer family, the recorded material entries were traversed in their original order and the first non-missing value was assigned for each property, producing descriptors such as `ETL_band_gap`, `HTL_HOMO` and `Perovskite_dielectric_constant`. Because the lookup was performed independently for each property, the material that supplies one descriptor is not necessarily the one that supplies another within a layer family, so a layer's descriptor vector can mix properties of different materials in a multi-layer stack. Missing values were retained and handled during model preparation (median imputation with a missingness indicator).

2.3.3. PDP-Derived Process and Device Features

The composition descriptors were combined with relevant device and fabrication variables retained from the original PDP export. These variables included available layer-thickness information, deposition methods and solvents, annealing conditions, and device-related flags. Together with the composition descriptors they form the PDP-derived tier $\mathcal{M}_1$ (Section 2.5.1). Because the parsed material identifiers were not used as predictors, $\mathcal{M}_1$ contains no explicit information on which transport-layer materials were used; that information enters the model only through the material-property descriptors of $\mathcal{M}_2$.

2.3.4. LLM-Derived Recipe Flags ($\mathcal{M}_3$)

The LLM did not emit the 36 recipe flags directly. They were built afterwards, deterministically and without new LLM calls or further full-text retrieval, from the stored extraction JSON by regular-expression matching of canonicalised material names and process fields. Keys that name photovoltaic outcomes were skipped during matching. The flags are binary and paper-level (`rc_*`) and are intended to encode recipe information that the PDP columns do not hold as explicit identities: 14 additive and precursor flags (for example chloride and thiocyanate sources, acetate, SnF_2, 5-AVA, hydrohalic acids, alkali and Cs additives, ammonium-salt and spacer cations, Lewis bases, polymer, water and DIO additives, carbon-based materials); 6 hole-transport-layer dopant flags (TBP, LiTFSI, FK-type dopants, F4-TCNQ, Cu salts, borates); 2 electron-transport-layer flags ($TiCl_4$ treatment, other dopants); 2 undoped-layer and interface-passivation flags; 7 process-route flags (antisolvent, vacuum, two-step, hot casting, solvent annealing, nitrogen box, ambient); and 5 device-context flags (flexible, tandem, lead-free, inorganic, mesoscopic). A flag is 1 if its pattern occurs in the non-outcome text of the publication's extracted material records and 0 otherwise, so it means "mentioned in the extraction" and not necessarily "used in every device of the paper".

The patterns are broad lexical matches and the flag names should be read accordingly. For example, the ammonium-salt flag (`rc_pea_family`, positive for 60.8% of extracted papers) includes the generic term "ammonium" and therefore also fires on methylammonium compounds: only 104 of its 1,723 positive papers contain a spacer-specific term (PEAI, BAI, OAI, GAI, NH_4I or phenethyl/butyl/octyl-ammonium). The carbon flag includes fullerene, PCBM and nanotube terms, and the Lewis-base flag includes pyridine and so fires on the TBP additive. One flag (hot casting) is positive for a single publication and is effectively constant. Coarse flags of this kind partly encode perovskite composition and transport-layer choice, which $\mathcal{M}_1$ and $\mathcal{M}_2$ already contain, and this is a plausible reason why their incremental value is small. The flags are missing for publications without an extraction record, which covers 15,027 of the 41,107 records (36.6%; 2,731 publications; 131 of these records, from 32 publications, have an empty extraction and all-zero flags). They are passed to CatBoost as native missing values without missingness indicators. $\mathcal{M}_3$ is $\mathcal{M}_2$ plus these 36 flags, which gives 215 predictor columns (179 of $\mathcal{M}_2$ plus 36), verified on the fitted models. In the $\mathcal{M}_3$ models a CatBoost first-feature-use penalty of 0.5 is applied to the flag columns and none to the other columns; this setting is specific to the added features and is therefore a difference between the $\mathcal{M}_3$ and $\mathcal{M}_2$ configurations (Table 3; its effect is examined in Section 3.2). Photovoltaic outcome fields were not used. The flags are noisy paper-level outputs, have not been validated against a manually annotated gold

set, and cannot distinguish devices within a publication. The earlier 44-predictor literature representation is a different feature set and is not a subset of $\mathcal{M}_3$.

2.4. Data Integration, Filtering and Feature Tiers

The structured device-level feature table and the DOI-level literature feature table were integrated using the publication DOI as the join key. DOI strings were normalised before matching and before grouping by removing byte-order marks and whitespace, lower-casing, and stripping `doi.org` and `doi:` prefixes; this merges case and prefix variants, so that the 7,499 distinct DOI strings of the raw export correspond to 7,454 publications and the 7,371 strings of the filtered data to 7,326 publications. A left join was performed from the device-level table to the DOI-level feature table, retaining device records even when no literature-feature record was available. Missing paper-level features arising from unsuccessful retrieval, incomplete extraction or unmatched DOIs were preserved as missing values.

The PDP export contained 43,398 records, of which 286 have no DOI. We excluded 1,002 records with missing target values and a further 1,289 records with reported PCE below 1.0% (3.0% of the records with a PCE value), which leaves 41,107 device records: records from 7,326 DOI-identified publications plus 232 records without a DOI (Table 1). Of these, 28,340 records are n-i-p and 12,684 are p-i-n devices; the remaining 83 have other or unspecified architectures and were used only in the pooled model. The exclusion of low-efficiency records is a selection step: the 1,289 removed records (median PCE 0.27%; 562 publications contribute at least one) are devices that are non-functional, poorly performing or possibly mis-entered, so the models and every score reported here describe devices with PCE $\geq 1\%$ and say nothing about the prediction of failed devices; the truncation also reduces the variance of the target and therefore affects R^2.

The filtered set contains 17 records (six publications) with PCE above 27%, above the certified single-junction record. The full PDP export, unlike the processed modelling table, records the illumination: all 17 were measured at reduced intensity (0.03 to 0.31 on the PDP scale where standard illumination is 100), twelve of them labelled "indoor light" and five without a spectrum label, so these values are not comparable with standard-illumination PCE. We retained them under the pre-declared filter and report a sensitivity analysis instead: across the filtered data, 41 records have a spectrum other than AM1.5 and 249 an intensity outside 90–110, and removing these records (which include the 17) from the *test* sets, without retraining, raises the pooled R^2 of $\mathcal{M}_1$, $\mathcal{M}_2$ and $\mathcal{M}_3$ on the reserved publications by about 0.005 to 0.006 and leaves the ordering and the size of the differences between tiers unchanged (Supplementary Table S9). The DOI was used exclusively as a record-linkage and grouping variable and was never a predictor. The 232 records without a DOI were treated as one explicit group that is never divided between partitions; this is conservative, because their true publications are unknown. The between-publication variance share in Table 1 is the one-way random-effects ANOVA intraclass correlation, ICC(1), computed with the adjusted mean group size for unbalanced groups, over the 7,327 groups (7,326 publications and the DOI-less block). Free-text columns were removed. All data-dependent preprocessing was fitted on the training partition of each split only: columns with more than 80% missing values were dropped, numerical features with missing entries were median-imputed and accompanied by a binary missingness indicator (suffix `_na`), and missing categorical features were assigned the label "unknown"; the recipe flags of $\mathcal{M}_3$ are the exception and are kept as native missing values. The predictor columns of each tier are listed in the Supplementary Information (Table S1).

2.5. Machine Learning Models and Evaluation

2.5.1. Prediction task and feature tiers.

We formulated PCE prediction as a supervised regression task, with experimentally reported PCE as the target. Three nested feature tiers are used throughout (Table 2): $\mathcal{M}_1$, the PDP-derived device, processing, architecture and composition features; $\mathcal{M}_2$, $\mathcal{M}_1$ plus curated material-property descriptors; and $\mathcal{M}_3$, $\mathcal{M}_2$ plus the 36 LLM-derived recipe flags (Section 2.3.4). Nesting is enforced in code and asserted in the test suite. The three tiers are compared on the same reserved publications (Section 2.5.4). The repeated-holdout, record-level and stricter protocols, the reference models and the tuned models (Section 3.3 onwards) were run for $\mathcal{M}_1$ and $\mathcal{M}_2$ only; they pre-date the recipe flags. In those experiments, the five-holdout results are also given separately for the p-i-n and n-i-p subsets, which were modelled independently, and an earlier

Table 1: Summary of the dataset used for modelling.

Quantity	Value
Records in PDP export (publications after DOI normalisation)	43,398 (7,454), of which 286 without DOI
Removed: missing PCE	1,002
Removed: PCE < 1.0%	1,289
Records after filtering (DOI-identified publications)	41,107 (7,326)
of which records without DOI	232
n-i-p / p-i-n / other architecture	28,340 / 12,684 / 83
Records with recipe flags, $\mathcal{M}_3$ (DOI-associated publications)	15,027 (2,731; 36.6%)
Records with the earlier literature representation (publications)	14,896 (2,699; 36.2%)
PCE after filtering: mean, median, range (%)	12.4, 13.0, 1.0–36.2
Records per publication: median, maximum	4, 127
Between-publication share of target variance (ICC(1))	0.69

44-predictor literature representation, which is not $\mathcal{M}_3$, was evaluated and is reported in the Supplementary Information.

Table 2: Feature tiers. Counts are predictor columns of the pooled model after preprocessing, including missingness indicators, as verified on the fitted models (in the five-holdout experiments the indicator set is derived from each training partition, so the n-i-p and p-i-n models contain on average 76/178 and 77/179 features for $\mathcal{M}_1/\mathcal{M}_2$). The 36 recipe flags of $\mathcal{M}_3$ are kept as native missing values and have no indicators. Exact lists: Supplementary Table S1.

Feature group	Tier	No. of features	Content
Process	$\mathcal{M}_1$	21	Deposition methods, annealing, layer thickness
Device	$\mathcal{M}_1$	9	Architecture and device flags
Composition	$\mathcal{M}_1$	47	A-, B- and X-site ion fractions and counts
Material properties	$\mathcal{M}_2$	102	Band gap, HOMO/LUMO, mobility, etc. (49 base columns plus 2 band-alignment differences and indicators)
Recipe flags (LLM)	$\mathcal{M}_3$	36	Binary paper-level flags: additive and precursor identities, ETL/HTL dopants, process route, device context
Total	$\mathcal{M}_1$ / $\mathcal{M}_2$ / $\mathcal{M}_3$	77 / 179 / 215	

2.5.2. Model training.

Models were trained with CatBoost[24], using its native handling of the categorical variables (architecture and the deposition methods). To ensure a controlled comparison across feature tiers, all tiers used a single CatBoost regressor with the same fixed hyperparameters (Table 3). These values were fixed a priori and inherited from an earlier configuration of this project that was not tuned in the present study; the number of trees was chosen by early stopping on a DOI-disjoint validation set and no tier was tuned. Apart from the predictor set, the only tier-specific setting is the first-feature-use penalty of 0.5 applied to the 36 recipe-flag columns of $\mathcal{M}_3$ (Table 3); the tier comparisons are therefore feature ablations with one added regularisation setting for the added columns, and not comparisons between model architectures, ensembling or tuned hyperparameters. Separately tuned configurations are used only where stated (Section 3.2) and are never mixed with the matched comparison. Reference models were trained on the same splits and features: a training-mean predictor; ridge regression (one-hot categorical encoding, standardised and winsorised numerical features, penalty chosen on the validation set); XGBoost with library defaults (the setting of Roberts-style pipelines); a random forest (300 trees); a multilayer perceptron (two hidden layers, early stopping on the validation set); and an exact Gaussian process on a fixed random subsample of 3,000 training records (exact GPR scales cubically with the number of records). A separate model-development

experiment, *not* used for the tier comparison, selected CatBoost depth and L2 regularisation from a six-point grid by validation RMSE (identical grid for every tier), then averaged three seeds, and combined this bag with an early-stopped XGBoost model.

Table 3: Matched CatBoost configuration used for all tiers, splits and protocols. The penalty row is the only tier-specific setting.

Setting	Value
Loss, learning rate, depth	RMSE, 0.05, 8
Maximum iterations, early stopping	1,500 (6,000 for the record-level arm, see text), 100 rounds on the validation set
L2 regularisation, row subsample, feature subsample per level	3.0, 0.8, 0.7
Minimum data in leaf, seed	5, 42
First-feature-use penalty	0.5 on the 36 recipe-flag columns of $\mathcal{M}_3$ only; 0 for all other columns and tiers

2.5.3. Evaluation protocol with a DOI-disjoint validation set.

To evaluate generalization to previously unseen publications, we used five repeated DOI-grouped holdout splits generated with `GroupShuffleSplit` (test fraction 0.2 of publications, seed 42), so that records sharing a normalised DOI were always assigned to the same partition (Table 4). Within each training partition, a further DOI-disjoint validation set (15% of the training publications) was used for early stopping and, where applicable, model selection; the test partition was used only for the final evaluation, and the model was not refitted on the validation records. No DOI occurs in more than one of the training, validation and test partitions in any split; this is recorded in every run record and checked by automated tests. Because repeated group-shuffle splits may have overlapping test sets, the evaluation is a repeated DOI-grouped holdout and not a mutually exclusive five-fold cross-validation. The standard deviations reported across splits describe split-to-split variation of the score; they are not confidence intervals, and the five splits are not independent replications. The record-level comparison protocol used `ShuffleSplit` with the same fractions, seed scheme, tiers, preprocessing and models; its validation set is a random subset of the training records. The record-level and DOI-grouped arms share the predictor definitions, model family, preprocessing and filtering, but they are not identical in training budget or in the composition of the validation set: record-level fits keep improving on a random validation set for much longer than DOI-grouped fits (a first run with the 1,500-iteration cap truncated 31 of 45 record-level CatBoost fits). The record-level arm was therefore rerun with a cap of 6,000 iterations for all subsets, and for the pooled data again with a cap of 20,000 iterations, where all 15 fits stopped on their own (best iteration 7,207–11,702); the pooled record-level numbers reported here are the converged ones. DOI-grouped fits stop before 500 iterations and are unaffected by the cap, and the validation set of the record-level arm is a random subset of training records and therefore shares publications with the training records. Two stricter protocols were run for the pooled and architecture-specific models with the same tiers, preprocessing, models and validation scheme. *Lead-author grouping* assigns each publication the modal PDP lead-author surname (a proxy for laboratory; common surnames such as Wang, Zhang and Li make some groups large and heterogeneous, so this over-groups) and splits by that key with the same `GroupShuffleSplit` settings; records without DOI remain one block. *Forward-in-time splitting* dates each publication by the earliest value of the PDP `Ref_publication_date` field among its records, a date held in the database that need not equal the journal publication date, and, for three non-overlapping test windows covering publication-date quantiles (0.55, 0.70], (0.70, 0.85] and (0.85, 1.00] (test dates 2018-05 to 2024-08), trains on strictly earlier publications, with the most recent 15% of those as validation set; all partitions are DOI-disjoint by construction. Paired differences between tiers were assessed within each split by a cluster bootstrap that resamples test publications with replacement (1,000 resamples) and reports the per-split 95% percentile interval. This bootstrap addresses dependence among the records of a test publication; it does not capture variation due to the choice of training publications or to model selection. Because the test sets

of different repeats overlap, no pooled significance test is reported.

Table 4: Data-splitting protocol (pooled dataset; means over five splits).

Item	Setting
Grouping variable	Normalised publication DOI; records without DOI form one block
Splitter	`GroupShuffleSplit`, 5 repeats, seed 42 (inner split seed 42+k)
Training set	about 27,700 records, 4,981 publications
Validation set (15% of training publications)	about 5,000 records, 880 publications; early stopping/model selection only
Test set (20% of publications)	about 8,430 records, 1,466 publications
DOI overlap, train/validation/test	Zero in every split and partition pair
Comparison protocol	Record-wise `ShuffleSplit`, same fractions and seeds
Reported statistics	Mean ± s.d. over the five splits

2.5.4. Common-split evaluation of the three tiers.

$\mathcal{M}_1$, $\mathcal{M}_2$ and $\mathcal{M}_3$ were compared under one protocol fixed after the exploratory ablation. Publications were assigned by a salted DOI hash to a development partition (65%; 26,982 records, 4,708 publications) used for all design, tuning and ablation work, and a reserved partition (35%; 14,125 records, 2,619 publications, of which 5,460 records from 1,005 publications have extraction data). The reserved publications had appeared in earlier experiments with other feature representations, so they are new for the recipe-flag representation but not untouched data, and the flag block was selected after comparisons on development publications. For the main comparison all three tiers used the matched default CatBoost configuration (Table 3), the same development records for training, the same DOI-disjoint 15% validation set per seed for early stopping, training-only preprocessing, and the same five seeds; the final prediction is the mean of the five seed models, and all tiers are scored on identical reserved records. Models were trained once on all development records. The $\mathcal{M}_2$ and $\mathcal{M}_3$ arms were scored first; the $\mathcal{M}_1$ arm was added afterwards in the same script path, with the same seeds and validation splits, and no choice depended on its result. Scores are reported for the covered subset (reserved records from publications with extraction data) and for all reserved records, where records of publications without extraction data carry missing flags under the missing-feature policy above. Paired differences between tiers were assessed by a publication-cluster bootstrap of the squared error (2,000 resamples), and per-tier R^2 intervals by the same bootstrap. To examine configuration dependence of the $\mathcal{M}_2 \to \mathcal{M}_3$ step only, an equal-budget random search of 24 CatBoost configurations (the matched default included) was run for $\mathcal{M}_2$ and $\mathcal{M}_3$ separately and scored by DOI-grouped five-fold mean squared error on development publications; $\mathcal{M}_1$ was not tuned. Two comparisons then isolate different questions: *same settings* gives both tiers the same configuration (the default, or one tier's tuned configuration), so that only the added features differ; *each tuned* gives each tier its own selected configuration, which mixes feature and tuning effects. Negative controls replaced the flag block by five DOI-permuted copies or by a coverage indicator alone. Three analyses use the saved predictions or one additional fit and are labelled as sensitivity analyses. (i) *Publication-weighted metrics* give each reserved record the weight $1/n_p$, where n_p is the number of reserved records of its publication within the population, so that every publication counts equally: weighted $R^2 = 1 - \sum_i w_i e_i^2 / \sum_i w_i (y_i - \bar{y}_w)^2$ with the weighted mean $\bar{y}_w$, and weighted MAE and RMSE are weighted means; paired intervals use the same publication-cluster bootstrap. (ii) *Illumination sensitivity* removes records with a non-AM1.5 spectrum or an intensity outside 90–110 from the test sets without retraining. (iii) *Unpenalised* $\mathcal{M}_3$: the $\mathcal{M}_3$ models were refitted once with the first-feature-use penalty set to 0, on the same development records, seeds, validation sets and default configuration, and scored on the same reserved records. The reserved publications were examined in earlier experiments, so this run is a robustness check and not independent confirmation. Because this partition and protocol differ from the five repeated holdouts below, absolute R^2 values from the two protocols are not comparable.

2.5.5. Performance metrics.

Predictive performance was assessed using the coefficient of determination (R^2), mean absolute error (MAE), and root mean squared error (RMSE), calculated by comparing experimentally reported PCE values, y_i, with predictions, $\hat{y}_i$, for the N records in each held-out test partition:

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}, \tag{3}$$

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|, \tag{4}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}, \tag{5}$$

where $\bar{y}$ is the mean experimentally reported PCE in the corresponding test partition. MAE and RMSE are expressed in PCE percentage points. The coefficient of determination measures performance relative to a constant-mean reference, whereas MAE and RMSE quantify absolute prediction errors, with RMSE assigning greater weight to large deviations.

2.6. Feature Importance and Model Interpretation

SHAP (Shapley additive explanations) was used to interpret fitted CatBoost models[25]. The SHAP analysis was computed for matched CatBoost models trained on $\mathcal{M}_2$ plus the earlier 44-predictor literature block, not for $\mathcal{M}_1$, $\mathcal{M}_2$ or $\mathcal{M}_3$ models. It used `TreeExplainer` on a random sample of 1,500 held-out records from the first DOI-grouped split, with mean absolute SHAP values as global importance, for the pooled, p-i-n and n-i-p models; features were aggregated into feature groups. Because the p-i-n and n-i-p models were trained on different records, differences between their rankings describe what each model learned rather than different physical effects in the two architectures.

3. Results and Discussion

3.1. Effect of the feature tiers on reserved publications

Adding the LLM-derived recipe flags improved the observed held-out predictions under matched settings (Table 5, Table 6 and Fig. 3). All three tiers were trained with the same CatBoost configuration, on the same development publications and early-stopping splits, and scored on identical reserved publications; the sequences below are therefore the observed effect of adding the predictors of each tier. On the 5,460 reserved records from 1,005 publications with extraction data, adding the flags to $\mathcal{M}_2$ raised R^2 from 0.359 to 0.366 ($\Delta R^2 = +0.0071$) and lowered MAE from 2.911 to 2.892 and RMSE from 3.747 to 3.726 percentage points. On all 14,125 reserved records from 2,619 publications, R^2 rose from 0.403 to 0.406 ($\Delta R^2 = +0.0025$), MAE fell from 2.955 to 2.949 and RMSE from 3.794 to 3.787. The change is positive in all three point metrics in both populations and is larger on the publications for which the flags exist, which is consistent with the information being carried by the flags; on all records, 61% of the reserved records have no flags and cannot benefit. The paired publication-cluster bootstrap intervals for ΔR^2 are -0.0007 to $+0.0157$ (covered) and -0.0011 to $+0.0062$ (all records); they include zero, which leaves the magnitude and the generalizability of this positive observed effect uncertain. The difference was positive in four of the five seed models in both populations; the seed models share the same training publications and are not independent replications, so the paired publication-cluster intervals are the relevant uncertainty statement.

The preceding step, from the PDP-derived baseline $\mathcal{M}_1$ to the material-property tier $\mathcal{M}_2$, is larger and its intervals exclude zero: $\Delta R^2 = +0.042$ ($+0.021$ to $+0.063$) on covered records and $+0.032$ ($+0.021$ to $+0.044$) on all records, positive in all five seed models, with MAE reductions of 0.110 and 0.099 percentage points. Across the three tiers R^2 rises from 0.318 to 0.359 and 0.366 on covered records and from 0.371 to 0.403

and 0.406 on all records, and the sequence is monotonic in R^2, MAE and RMSE in both populations. The per-tier intervals in Table 5 overlap widely because they are dominated by between-publication variability shared by all tiers; the paired intervals of Table 6 are the appropriate ones for differences. R^2 on the covered subset is lower than on all records for every tier because it is a different population with a different PCE variance, not because of the flags. The material-property step is examined further in Section 3.6, where a material-identity control shows that most of it can be attributed to knowing which materials are present.

Table 5: Common-split comparison of the three tiers on reserved publications (matched default CatBoost for all tiers; mean of five seed models; trained on development publications only). R^2 is given with its 95% publication-cluster bootstrap interval; MAE and RMSE in PCE percentage points. LLM-covered: reserved records from publications with extraction data. The test records are identical for the three tiers within each population.

Population	Tier	Test records	Test DOIs	R^2	MAE	RMSE
LLM-covered	$\mathcal{M}_1$	5,460	1,005	0.318 [0.265, 0.365]	3.021	3.866
LLM-covered	$\mathcal{M}_2$	5,460	1,005	0.359 [0.304, 0.408]	2.911	3.747
LLM-covered	$\mathcal{M}_3$	5,460	1,005	0.366 [0.316, 0.412]	2.892	3.726
All reserved	$\mathcal{M}_1$	14,125	2,619	0.371 [0.337, 0.405]	3.054	3.896
All reserved	$\mathcal{M}_2$	14,125	2,619	0.403 [0.370, 0.435]	2.955	3.794
All reserved	$\mathcal{M}_3$	14,125	2,619	0.406 [0.372, 0.438]	2.949	3.787

Table 6: Paired differences between tiers on the reserved publications of Table 5. ΔR^2 with a 95% paired publication-cluster bootstrap interval; ΔMAE and ΔRMSE are differences of the point metrics of Table 5 (percentage points; negative is better); "Seeds > 0" counts the five seed-level ΔR^2 that are positive.

Population	Comparison	ΔR^2 [95% interval]	ΔMAE	ΔRMSE	Seeds > 0
LLM-covered	$\mathcal{M}_2 - \mathcal{M}_1$	+0.0416 [+0.0208, +0.0631]	−0.110	−0.120	5/5
LLM-covered	$\mathcal{M}_3 - \mathcal{M}_2$	+0.0071 [−0.0007, +0.0157]	−0.019	−0.021	4/5
LLM-covered	$\mathcal{M}_3 - \mathcal{M}_1$	+0.0487 [+0.0293, +0.0691]	−0.129	−0.140	5/5
All reserved	$\mathcal{M}_2 - \mathcal{M}_1$	+0.0323 [+0.0207, +0.0442]	−0.099	−0.101	5/5
All reserved	$\mathcal{M}_3 - \mathcal{M}_2$	+0.0025 [−0.0011, +0.0062]	−0.005	−0.008	4/5
All reserved	$\mathcal{M}_3 - \mathcal{M}_1$	+0.0348 [+0.0229, +0.0468]	−0.105	−0.109	5/5

3.2. Recipe flags: controls, settings and sensitivity analyses

The recipe-flag block was developed on development publications only. There, the remaining content of the extraction JSON (token bags, numeric quantities, device-matched layer values, gap-filling of PDP fields) added at most about +0.004 R^2 to $\mathcal{M}_2$, and giving $\mathcal{M}_2$ the PDP's own curated process columns for the same papers added only +0.022, which motivated a block of recipe identities. This comparison is itself a database-only reference: the PDP additive, solvent and quenching columns are curated device-level fields that lie outside the restricted $\mathcal{M}_1/\mathcal{M}_2$ definition, and supplying them raised development R^2 by more than the flags did (additives +0.012, solvents and quenching +0.003). It is not a like-for-like baseline for the flags, which are text-mined paper-level identities, and we did not build a non-LLM text-mining baseline. We therefore do not claim that the flags contain information absent from the PDP; the absence of such a baseline is a limitation. On covered development publications the flags gave +0.007 R^2 (12 of 15 folds), five DOI-permuted copies gave −0.009 to +0.002, and dropping any one flag group reduced the gain without any single group reproducing it, so the signal is distributed across the flags.

On the reserved publications, the observed gain is above that of every negative control. Under the default settings, five DOI-permuted copies of the flag block, which keep the marginals and the availability mask but remove the association with the publication, gave ΔR^2 from −0.0049 to +0.0015 (covered) and from −0.0028 to +0.0007 (all records), and a coverage indicator alone gave +0.0015 and +0.0011; the flags gave +0.0071 and +0.0025 (Fig. 4). The flags are also not only a coverage effect, since the indicator control is smaller.

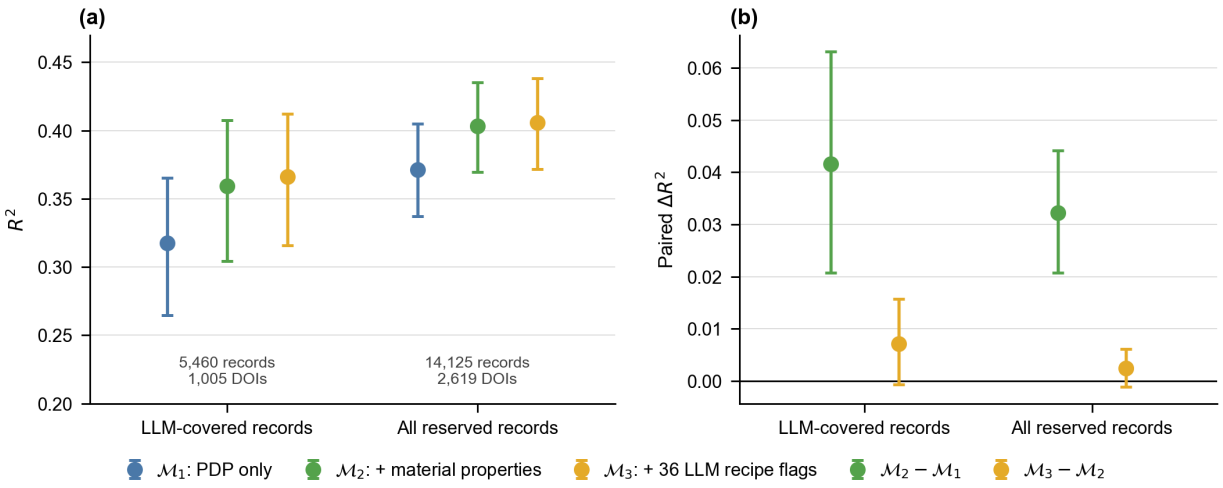


Figure 3: Common-split comparison of $\mathcal{M}_1$, $\mathcal{M}_2$ and $\mathcal{M}_3$ on reserved publications (matched default CatBoost). (a) R^2 per tier with 95% publication-cluster bootstrap intervals; (b) paired differences $\mathcal{M}_2 - \mathcal{M}_1$ and $\mathcal{M}_3 - \mathcal{M}_2$ with 95% paired bootstrap intervals, for LLM-covered records (5,460 records, 1,005 DOIs) and all reserved records (14,125 records, 2,619 DOIs). The $\mathcal{M}_3 - \mathcal{M}_2$ intervals include zero.

Table 7 shows the $\mathcal{M}_2 \rightarrow \mathcal{M}_3$ step under other estimator settings. When each tier used its own tuned configuration, covered-record R^2 was 0.361 for $\mathcal{M}_2$ and 0.371 for $\mathcal{M}_3$ ($\Delta R^2 = +0.0094$, -0.0016 to $+0.0219$; 5 of 5 seeds), and on all records both tiers reached about 0.41 ($\Delta R^2 = +0.0015$, -0.0039 to $+0.0067$). This tuned comparison is secondary and separate from the three-tier sequence: $\mathcal{M}_1$ was not tuned under this protocol, tuning alone moved $\mathcal{M}_2$ by $+0.0020$ (covered) to $+0.0030$ (all records), which is comparable to the flag effect, and the tuned arms mix feature and tuning effects, so it is not used to attribute the whole gain to the flags. With $\mathcal{M}_2$'s tuned configuration applied to both tiers the covered ΔR^2 is $+0.0024$ and the all-record difference -0.0016, so the size of the effect depends on the estimator configuration.

Two further sensitivity analyses leave the picture unchanged. Publication-weighting, which gives every publication the same total weight, yields $\mathcal{M}_2 \rightarrow \mathcal{M}_3$ changes of $\Delta R^2 = +0.0040$ (-0.0020 to $+0.0100$) on covered and $+0.0016$ (-0.0012 to $+0.0045$) on all reserved records, against $+0.0071$ and $+0.0025$ with record weights (Supplementary Table S10). Publication-weighted R^2 is higher than the record-weighted value for every tier (for example 0.437 against 0.406 for $\mathcal{M}_3$ on all records) because publications that contribute many device rows are somewhat harder to predict, and the $\mathcal{M}_1 \rightarrow \mathcal{M}_2$ step is $+0.037$ (all) and $+0.048$ (covered) under publication weights. Refitting $\mathcal{M}_3$ without the first-feature-use penalty on the flags (the other settings unchanged) gives $\Delta R^2 = +0.0089$ ($+0.0008$ to $+0.0177$) on covered and $+0.0034$ (-0.0004 to $+0.0072$) on all records, with MAE 2.885 and 2.944; the covered interval lies just above zero. Because this run was added after the reserved publications had already been examined, and is one of several settings that were looked at, we treat it as showing that the penalty is not what produces the gain and not as a confirmatory result; on development folds the penalties 0, 0.5 and 2 gave $+0.0052$, $+0.0070$ and $+0.0038$. Taken together, the observed effect of the flags is consistently positive in point estimate across controls, weightings and settings, small in size, and not established by any pre-specified interval. The flags are available for 36.6% of records, which dilutes the benefit on the full data; larger coverage from full-text extraction of the roughly 4,500 publications that currently lack it would be needed to estimate the effect more precisely, and we did not do this. Extraction coverage is not random with respect to the data (Supplementary Table S11). Covered and uncovered records have similar PCE distributions (mean 12.3% and 12.5%, median 12.9% and 13.1%; 2.6% and 3.6% of records at or above 20%) and the same architecture mix (69% n-i-p and 31% p-i-n in both), but the share of records with an extraction record peaks at 45% for 2018 and falls to 9% for 2020 and to zero from 2021, so the flags describe the 2014–2019 literature and are absent for the most recent publications. A plausible reason for the small size of the effect is redundancy: the extracted numerical parameters largely

duplicate PDP columns already in $\mathcal{M}_1$, and the coarse flags partly restate composition and transport-layer choices (Section 2.3.4); we did not test this explanation. The results in Table 7 use the reserved partition of Section 3.1 and must not be combined with the five-holdout tables below.

Table 7: Dependence of the recipe-flag step ($\mathcal{M}_3$ versus $\mathcal{M}_2$) on the CatBoost configuration, on the reserved publications of Table 5. ΔR^2 is $\mathcal{M}_3$ minus $\mathcal{M}_2$ with a 95% publication-cluster bootstrap interval; "Seeds > 0" counts the five seed-level differences. *Default*: matched default configuration for both tiers (same as Table 5); *each tuned*: each tier's own selected configuration; $\mathcal{M}_3$*'s* and $\mathcal{M}_2$*'s* tuned configuration: that configuration applied to both tiers. Covered: 5,460 records from 1,005 publications with extraction data; all: 14,125 records from 2,619 publications.

Records	Settings	R^2 ($\mathcal{M}_2$)	R^2 ($\mathcal{M}_3$)	ΔR^2	95% interval	Seeds > 0
Covered	Default	0.359	0.366	+0.0071	[−0.0007, +0.0157]	4/5
Covered	Each tuned	0.361	0.371	+0.0094	[−0.0016, +0.0219]	5/5
Covered	$\mathcal{M}_3$'s tuned config.	0.364	0.371	+0.0066	[−0.0021, +0.0162]	4/5
Covered	$\mathcal{M}_2$'s tuned config.	0.361	0.364	+0.0024	[−0.0077, +0.0131]	3/5
All	Default	0.403	0.406	+0.0025	[−0.0011, +0.0062]	4/5
All	Each tuned	0.406	0.408	+0.0015	[−0.0039, +0.0067]	3/5
All	$\mathcal{M}_3$'s tuned config.	0.405	0.408	+0.0026	[−0.0015, +0.0067]	2/5
All	$\mathcal{M}_2$'s tuned config.	0.406	0.405	−0.0016	[−0.0064, +0.0030]	2/5

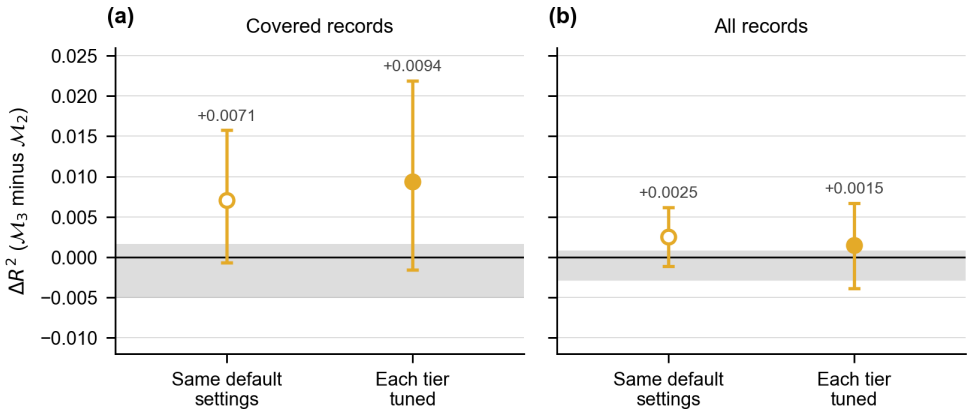


Figure 4: Effect of the recipe-flag block on reserved publications: ΔR^2 ($\mathcal{M}_3$ minus $\mathcal{M}_2$) with 95% publication-cluster bootstrap intervals for (a) covered records and (b) all records, under the same default settings (open symbols) and with each tier separately tuned (filled symbols). Grey bands mark the range of five DOI-permuted-block controls under default settings. All intervals include zero.

3.3. Generalization to unseen publications: repeated holdouts

The remaining experiments use five repeated DOI-grouped holdouts (Section 2.5.2) and the tiers $\mathcal{M}_1$ and $\mathcal{M}_2$; they pre-date the recipe flags and use a different protocol from Section 3.1, so their absolute values are not comparable with it. With the matched CatBoost configuration and $\mathcal{M}_2$, the pooled model reached $R^2 = 0.408 \pm 0.032$, MAE $= 2.95 \pm 0.08$ and RMSE $= 3.78 \pm 0.09$ percentage points on publications it had not seen (Table 8, Fig. 5; mean ± s.d. over five repeated DOI-grouped holdouts, each with a DOI-disjoint validation set for early stopping). The training-mean predictor has $R^2 \approx 0$ and MAE 4.08, so the model removes about 28% of the mean absolute error. Performance differed by architecture: R^2 was 0.404 ± 0.013 for n-i-p and 0.291 ± 0.065 for p-i-n devices, whereas MAE (2.97 and 3.05) and RMSE (3.83 and 3.96) were similar. The p-i-n result is also the less certain one, because the subset is smaller (about 2,600 test records and 451 test publications per split). Relative to the training-mean predictor, the p-i-n model reduces MAE

by 21% (3.84 to 3.05) and the n-i-p model by 28% (4.11 to 2.97), so on this data p-i-n devices are somewhat harder to predict, in addition to R^2 depending on the PCE variance within each subset.

Table 8: Performance of $\mathcal{M}_1$ and $\mathcal{M}_2$ with the same CatBoost configuration (Table 3) under repeated DOI-grouped holdout evaluation (mean ± s.d. over five splits; MAE and RMSE in PCE percentage points; N is the mean number of test records per split; the number of features varies slightly between splits because the missingness indicators are derived from each training partition).

Subset	Feature tier	N	Feat.	R^2	MAE	RMSE
Pooled	Database only ($\mathcal{M}_1$)	8,428	77	0.368 ± 0.024	3.081 ± 0.040	3.908 ± 0.046
Pooled	+ material ($\mathcal{M}_2$)	8,428	179	0.408 ± 0.032	2.950 ± 0.080	3.783 ± 0.085
p-i-n	Database only ($\mathcal{M}_1$)	2,598	77	0.244 ± 0.058	3.168 ± 0.049	4.085 ± 0.114
p-i-n	+ material ($\mathcal{M}_2$)	2,598	179	0.291 ± 0.065	3.047 ± 0.045	3.955 ± 0.131
n-i-p	Database only ($\mathcal{M}_1$)	5,612	76	0.374 ± 0.016	3.062 ± 0.047	3.918 ± 0.061
n-i-p	+ material ($\mathcal{M}_2$)	5,612	178	0.404 ± 0.013	2.966 ± 0.036	3.825 ± 0.060

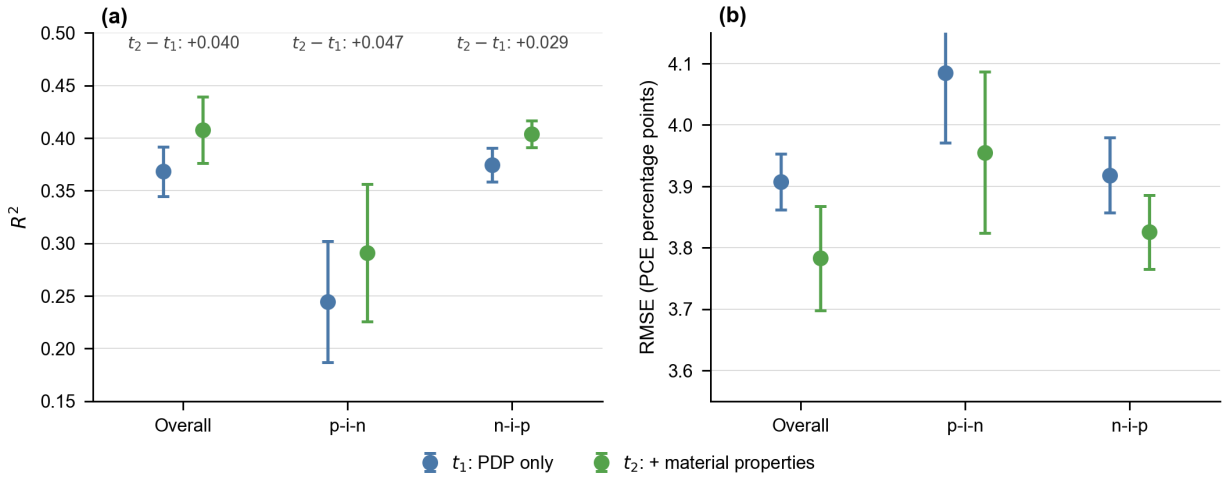


Figure 5: Effect of feature tier on prediction of unseen publications ($\mathcal{M}_1$ and $\mathcal{M}_2$; matched CatBoost, DOI-grouped splits). (a) R^2 and (b) RMSE, mean ± s.d. over five holdout splits; the vertical axes are truncated to the plotted range. The $\mathcal{M}_1 \rightarrow \mathcal{M}_2$ step is visible in all subsets. $\mathcal{M}_3$ was not part of this experiment; see Fig. 3.

The pooled results should be read against simple references (Table 9). Ridge regression on the same $\mathcal{M}_1$ features reaches $R^2 = 0.307 \pm 0.030$ and XGBoost with library defaults, a random forest, a validation-early-stopped MLP and a subsample-trained Gaussian process lie between 0.295 and 0.343. Matched CatBoost is the best of these by 0.025 (MLP) to 0.073 (random forest) in R^2; the margin over the MLP is comparable to the split-to-split standard deviation (0.02–0.03), so we do not claim a ranking among the non-linear models. A properly trained MLP and a Gaussian process are thus competitive with the tree ensemble on this dataset. The reference models other than ridge were run for $\mathcal{M}_1$ and for $\mathcal{M}_2$ plus the earlier literature block only (Supplementary Information), where CatBoost is likewise the best.

3.3.1. Model development: strongest validated predictor.

Separately from the controlled ablation, we asked how much a validation-tuned and ensembled model improves on the matched configuration (Table 10). For each split and tier, CatBoost depth and L2 regularisation were selected from a six-point grid by validation RMSE (the same grid and budget for every tier), the selected configuration was averaged over three seeds, and the result was averaged with an early-stopped XGBoost model. The ensemble reached $R^2 = 0.416 \pm 0.026$, MAE = 2.92 and RMSE = 3.76 percentage

Table 9: Reference models on the pooled dataset under the same five DOI-grouped splits (mean ± s.d.; MAE and RMSE in PCE percentage points). All models use validation data only for early stopping or penalty selection; XGBoost (defaults) and the random forest use no validation data. The Gaussian process is fitted to 3,000 randomly chosen training records. XGBoost, random forest, MLP and GPR were run for $\mathcal{M}_1$ only among the tiers of this paper.

Model	Tier	R^2	MAE	RMSE
Training-mean predictor	$\mathcal{M}_1$	-0.001 ± 0.001	4.079 ± 0.036	4.920 ± 0.047
Ridge regression	$\mathcal{M}_1$	0.307 ± 0.030	3.271 ± 0.039	4.093 ± 0.064
Ridge regression	$\mathcal{M}_2$	0.354 ± 0.032	3.130 ± 0.051	3.952 ± 0.075
CatBoost (matched)	$\mathcal{M}_1$	0.368 ± 0.024	3.081 ± 0.040	3.908 ± 0.046
CatBoost (matched)	$\mathcal{M}_2$	0.408 ± 0.032	2.950 ± 0.080	3.783 ± 0.085
XGBoost (library defaults)	$\mathcal{M}_1$	0.334 ± 0.031	3.148 ± 0.052	4.011 ± 0.070
Random forest	$\mathcal{M}_1$	0.295 ± 0.018	3.219 ± 0.041	4.128 ± 0.058
MLP (val. early stopping)	$\mathcal{M}_1$	0.343 ± 0.020	3.144 ± 0.029	3.987 ± 0.025
GPR (3,000-record subsample)	$\mathcal{M}_1$	0.330 ± 0.024	3.197 ± 0.039	4.025 ± 0.053

points for the pooled $\mathcal{M}_2$ data (p-i-n 0.300 ± 0.072, n-i-p 0.413 ± 0.013), 0.008 above the matched CatBoost. This gain comes from tuning, bagging and ensembling and not from any additional feature: the same pipeline gives 0.377 and 0.416 for $\mathcal{M}_1$ and $\mathcal{M}_2$, so the material-property step (+0.039) is again the measurable feature effect. This experiment was not run for $\mathcal{M}_3$ (and its results neither replace nor can be merged into Table 5 or Table 7); the separately tuned CatBoost models of Table 7 use a different partition and no bagging or ensembling. We regard $R^2 \approx 0.42$ and MAE ≈ 2.9 percentage points as the performance of a tuned $\mathcal{M}_2$ model on unseen publications supported by this evaluation, with the caveat that the split-to-split standard deviation is 0.03 and the validation-selected configuration changed between splits.

Table 10: Model-development experiment (DOI-grouped splits; R^2, mean ± s.d. over five splits). Tuned: CatBoost with depth/L2 chosen on the validation set; Bag: three-seed average of the selected configuration; XGB: early-stopped XGBoost; Ens: mean of Bag and XGB. This experiment differs from the matched ablation in estimator and budget, so it must not be used to attribute differences to features, and it is not the same experiment as the common-split evaluation (Tables 5 and 7); no $\mathcal{M}_3$ entries exist.

Subset	Feature tier	Tuned	Bag	XGB	Ens
Pooled	Database only ($\mathcal{M}_1$)	0.371 ± 0.023	0.372 ± 0.023	0.368 ± 0.019	0.377 ± 0.021
Pooled	+ material ($\mathcal{M}_2$)	0.408 ± 0.029	0.410 ± 0.029	0.409 ± 0.022	0.416 ± 0.026
p-i-n	Database only ($\mathcal{M}_1$)	0.235 ± 0.061	0.241 ± 0.060	0.235 ± 0.058	0.249 ± 0.059
p-i-n	+ material ($\mathcal{M}_2$)	0.293 ± 0.066	0.297 ± 0.069	0.285 ± 0.074	0.300 ± 0.072
n-i-p	Database only ($\mathcal{M}_1$)	0.374 ± 0.016	0.377 ± 0.015	0.368 ± 0.017	0.380 ± 0.015
n-i-p	+ material ($\mathcal{M}_2$)	0.406 ± 0.018	0.409 ± 0.017	0.402 ± 0.011	0.413 ± 0.013

The predicted-PCE distributions (Fig. 6) show that all models compress predictions toward the centre of the experimental distribution. Among the 8,224 test records of the first split, 272 reach 20% PCE or more; for the $\mathcal{M}_1$ models of that figure the spread of the predictions is 0.59, 0.60 and 0.59 times that of the experimental values (CatBoost, MLP, GPR), the maximum predictions are 21.6, 19.7 and 19.4%, and for CatBoost the test devices with PCE $\geq 20\%$ were under-predicted by 5.5 percentage points on average. For matched CatBoost with $\mathcal{M}_2$, the spread ratio on that split is 0.63, the maximum prediction 21.5%, and test devices with PCE $\geq 20\%$ were under-predicted by 5.1 points on average (4.8 points over all 1,328 such test records in the five splits). Regressors trained with a squared-error loss estimate conditional means, so extreme values regress toward the centre. We did not evaluate screening or ranking performance, and with moderate R^2 and this compression the models should not be used to identify record-efficiency devices or to guide experimental optimisation without such evidence. Six of 42,142 test predictions ($\mathcal{M}_2$, five splits, pooled) were negative (minimum -1.4%), which is unphysical and indicates that bounding predictions at zero is appropriate in use. The 17 records with PCE above 27% (six publications, up to 36.2%) were measured at reduced light intensity, twelve under indoor light (Section 2.4); excluding them from the test

sets changes the pooled $\mathcal{M}_2$ R^2 by only +0.003. Parity and residual plots (Fig. 7) show the same shrinkage as a residual that is negative at high PCE and positive at low PCE.

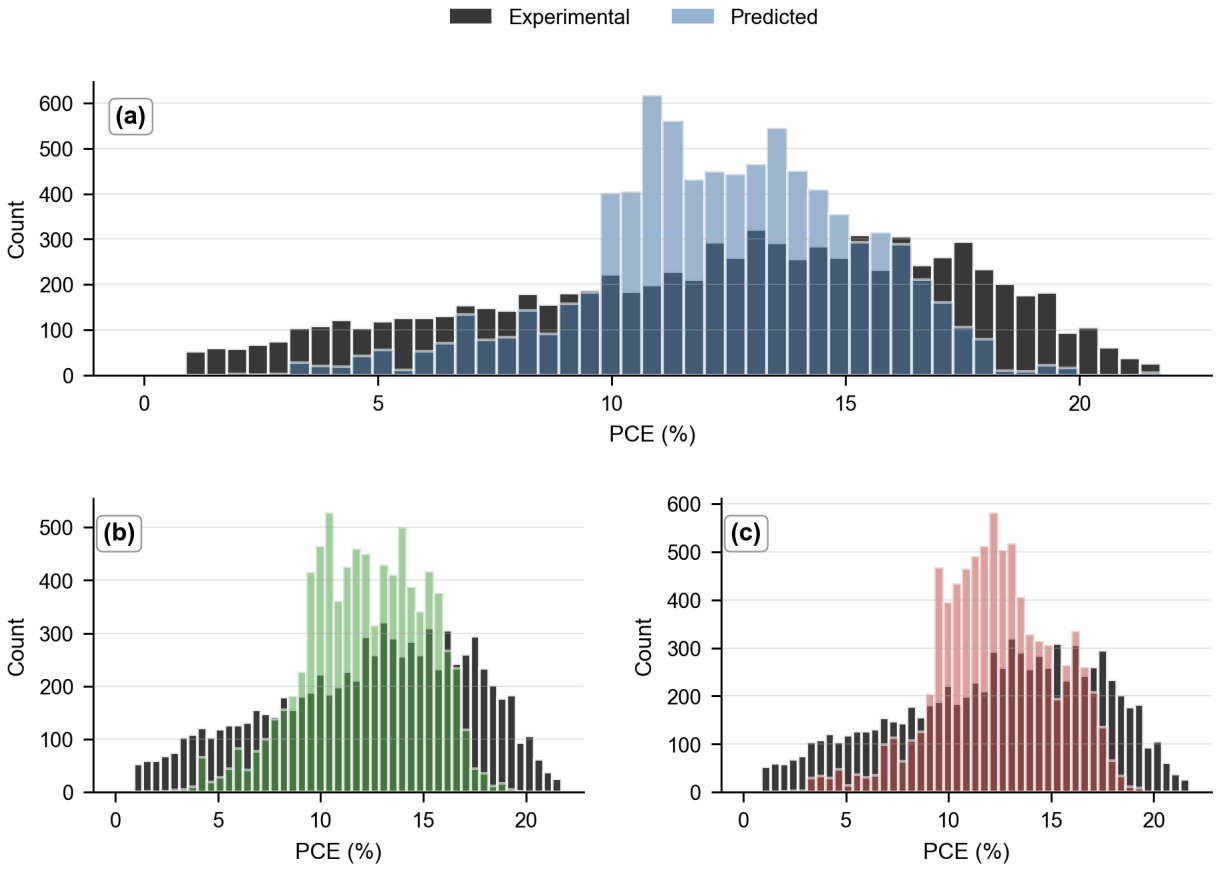


Figure 6: Distribution of experimental and predicted PCE on the first DOI-grouped split (pooled, $\mathcal{M}_1$) for (a) CatBoost, (b) the early-stopped MLP and (c) the subsample Gaussian process; $\mathcal{M}_1$ is the tier for which all three models were run. Axes extend to the 99.5th percentile of experimental PCE; 272 of the 8,224 test records reach 20% or more.

3.4. Effect of the evaluation protocol

The evaluation protocol had a large influence on measured performance. Table 11 and Fig. 8 compare record-level random splitting with DOI-grouped splitting across the feature tiers with identical features, preprocessing, model configurations, filtering and training budgets; only the unit that is split differs. For matched CatBoost the pooled R^2 was 0.587 and 0.618 under record-level splitting for $\mathcal{M}_1$ and $\mathcal{M}_2$, compared with 0.368 and 0.408 under DOI-grouped splitting, a difference of 0.219 and 0.210. MAE rose from 2.23–2.34 to 2.95–3.08 percentage points. The gap is therefore similar for both tiers. The two arms share predictor definitions, model family and preprocessing but not convergence behaviour: the record-level fits needed 7,200–11,700 iterations (DOI-grouped fits stop before 500), and the gap was 0.005–0.007 smaller when the cap was 6,000 iterations, so it is sensitive to training length but not to the cap once converged.

Two properties of the data make a large gap plausible. First, 69% of the variance of PCE lies between publications (one-way intraclass correlation by DOI, 0.69), with a median of four records per publication. Second, for the earlier representation of $\mathcal{M}_2$ plus the 44-predictor literature block (the only one for which this audit was run), 31,591 of 41,107 records (77%) share their complete predictor vector with at least one other record, but only 162 of 7,733 such groups span more than one publication, so near-duplicate devices

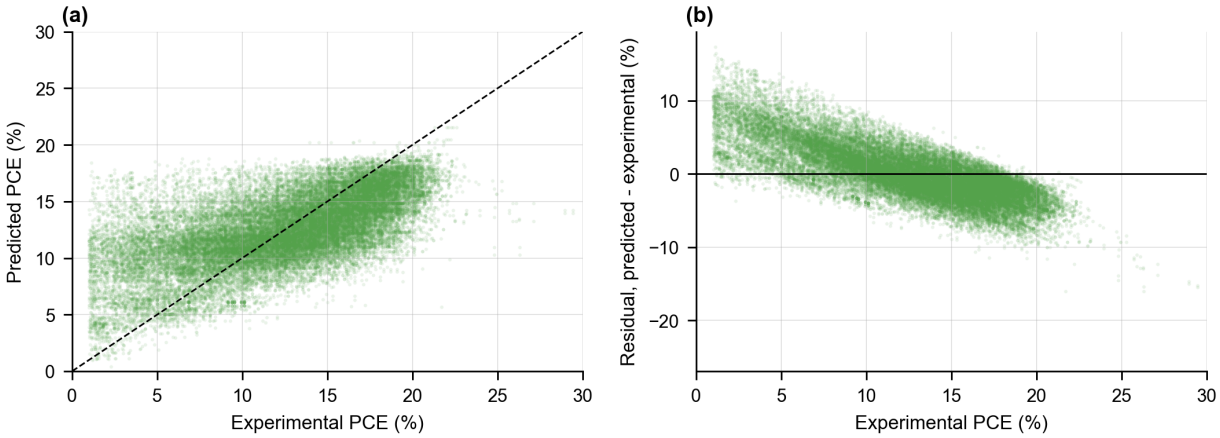


Figure 7: Matched CatBoost ($\mathcal{M}_2$, pooled): (a) parity and (b) residuals for the test records of the five DOI-grouped splits stacked (42,142 predictions; records may recur across splits because test sets overlap). The dashed line is $y = x$.

Table 11: Record-level random versus DOI-grouped splitting for the pooled dataset (mean ± s.d. over five splits with 20% of records or publications held out; MAE in PCE percentage points). Features, preprocessing, model configuration and filtering are identical under both protocols; the gap is the mean per-split difference in R^2.

Model	Feature tier	R^2 random	R^2 DOI-grouped	Gap	MAE rand.	MAE DOI
CatBoost	Database only ($\mathcal{M}_1$)	0.587 ± 0.009	0.368 ± 0.024	0.219	2.34	3.08
CatBoost	+ material ($\mathcal{M}_2$)	0.618 ± 0.006	0.408 ± 0.032	0.210	2.23	2.95
Ridge	Database only ($\mathcal{M}_1$)	0.363 ± 0.008	0.307 ± 0.030	0.056	3.09	3.27
Ridge	+ material ($\mathcal{M}_2$)	0.408 ± 0.008	0.354 ± 0.032	0.055	2.95	3.13

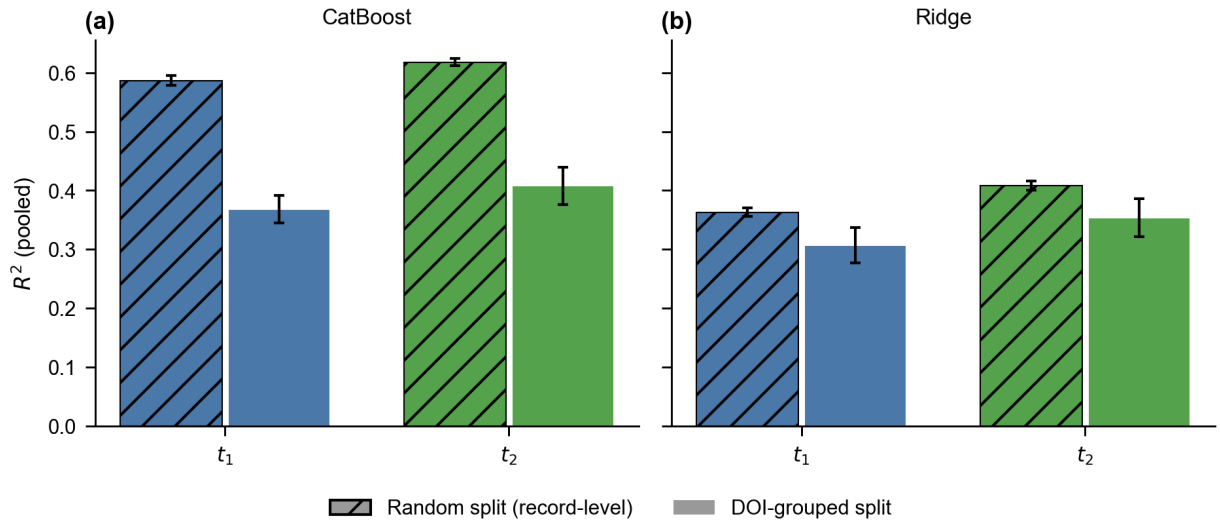


Figure 8: Record-level random (hatched) versus DOI-grouped (solid) splits for $\mathcal{M}_1$ and $\mathcal{M}_2$ with (a) CatBoost and (b) ridge regression (pooled dataset; mean ± s.d. over five splits).

are almost entirely within-publication; their PCE values differ by a median standard deviation of 1.2 points within a group. A flexible model can memorise publication-specific patterns that a random split leaves available at test time, and the size of the gap and the much longer training of the record-level fits are consistent with such publication-specific memorisation. The linear model fits this reading: for the pooled data ridge regression shows a much smaller gap (0.055–0.056), but it is not a clean capacity test, because the ridge gap for the p-i-n subset is large (0.11–0.18) and ridge is also a weaker predictor on all splits; we did not run a matched-capacity control. The comparison quantifies the sensitivity of the measured score to the splitting protocol on this dataset. It does not attribute every point of the gap to leakage of identical devices: the two protocols also produce test sets of different composition, and the record-level arm differs in convergence and validation composition. The stricter protocols of Section 3.5 show how much of the DOI-grouped score remains under lead-author grouping and forward-in-time evaluation, and none of them is an external test. DOI-grouped evaluation answers a specific question, namely performance on papers not seen in training. It does not remove similarity between papers from the same laboratory, and it is not a prospective or external validation. Earlier PDP studies report a larger drop on newly registered records (Fukasawa et al.: R^2 0.7177 under an 8:2 split to 0.45 on 294 newer records)[11], which is directionally consistent, although the targets, feature sets, dataset versions and test sets differ and the numbers are not comparable.

3.5. Stricter validation: laboratory proxy and forward-in-time splits

DOI-grouped splitting leaves two obvious routes for residual similarity: papers from the same laboratory and papers close in time. Table 12 compares both with DOI-grouped and record-level evaluation (pooled data, all models and tiers matched). Grouping by lead-author surname changes the CatBoost $\mathcal{M}_2$ R^2 from 0.408 to 0.394 ± 0.018 (MAE 2.92). This is the change produced by this coarse proxy, which both over-groups common surnames and misses laboratories whose lead authors differ; it should not be read as a measure of the size of the same-laboratory effect. Forward-in-time evaluation lowers the score more: training on records dated earlier and testing on later ones gives $R^2 = 0.345 \pm 0.049$ for $\mathcal{M}_2$ (MAE 3.18; individual windows 0.335, 0.398 and 0.302), against -0.155 for the training-mean predictor, whose negative R^2 reflects a shift of the PCE distribution towards higher efficiencies over time. Ridge regression degrades similarly (0.286). Neither protocol changes the material-property conclusion: the $\mathcal{M}_1 \rightarrow \mathcal{M}_2$ step is +0.030 (lead-author) and +0.052 (forward in time) in R^2 (per-window values in Table 13). MAE rises from 2.95 under DOI-grouped splitting to 3.18, while the MAE of the training-mean predictor rises from 4.08 to 4.57 on the later test windows. The temporal result is based on three windows of one dataset snapshot with a partly overlapping training history, so its spread should not be over-interpreted. It shows that the score depends on the time period of the test publications, which includes distribution drift toward higher efficiencies, and it is not an external validation. These protocols were not run for $\mathcal{M}_3$.

3.6. Contribution of material-property descriptors

Adding material-property descriptors ($\mathcal{M}_1 \rightarrow \mathcal{M}_2$) improved held-out predictions. In the five-holdout experiment R^2 increased by 0.040 (pooled), 0.047 (p-i-n) and 0.029 (n-i-p) and RMSE fell by 0.125, 0.130 and 0.093 percentage points (Table 14), consistent with the larger gain on the reserved publications (Table 6). The pooled gain had a cluster-bootstrap interval excluding zero in all five splits (two of five for p-i-n and three of five for n-i-p, which have fewer test publications), and R^2 improved in all 15 split–subset pairs. This is an empirical finding about the feature sets and should be distinguished from a stronger interpretation, namely that the physical property values supply independent transferable information. Because $\mathcal{M}_1$ contains no explicit transport-layer material identities (Section 2.3.3) and each property is a fixed function of the material, the $\mathcal{M}_2$ descriptors also identify which materials were used. A separate material-identity control, run as its own experiment (disjoint DOI-grouped five-fold, three seeds, the same CatBoost configuration, 41,107 records) and not part of the reserved-split table, adds only the ETL, HTL and perovskite identities to $\mathcal{M}_1$ as native CatBoost categorical features. It gives $R^2 = 0.406$, against 0.370 for $\mathcal{M}_1$ and 0.407 for $\mathcal{M}_2$: the 102 descriptor columns add +0.001 over identity alone (the interval of the paired MSE difference includes zero), and descriptors on top of identity add +0.004 (also not distinguishable from zero). Knowing which transport and absorber materials a device contains therefore accounts for essentially all of the observed

Table 12: Pooled R^2 (mean ± s.d. over splits) by validation protocol, model and feature tier, with matched features, preprocessing and model settings. Splits: five repeats (record-level, DOI-grouped, lead-author-grouped) or three forward windows. MAE refers to $\mathcal{M}_2$ (percentage points). The pooled record-level CatBoost fits are the converged ones (20,000-iteration cap; Section 3.4).

Protocol	Model	N_{test}	Splits	$\mathcal{M}_1$	$\mathcal{M}_2$	MAE ($\mathcal{M}_2$)
Record-level	CatBoost	8,222	5	0.587 ± 0.009	0.618 ± 0.006	2.23
Record-level	Ridge	8,222	5	0.363 ± 0.008	0.408 ± 0.008	2.95
Record-level	Mean	8,222	5	-0.000 ± 0.000	-0.000 ± 0.000	4.06
DOI-grouped	CatBoost	8,428	5	0.368 ± 0.024	0.408 ± 0.032	2.95
DOI-grouped	Ridge	8,428	5	0.307 ± 0.030	0.354 ± 0.032	3.13
DOI-grouped	Mean	8,428	5	-0.001 ± 0.001	-0.001 ± 0.001	4.08
Lead-author-grouped	CatBoost	8,280	5	0.363 ± 0.019	0.394 ± 0.018	2.92
Lead-author-grouped	Ridge	8,280	5	0.313 ± 0.011	0.353 ± 0.017	3.05
Lead-author-grouped	Mean	8,280	5	-0.002 ± 0.002	-0.002 ± 0.002	3.98
Forward in time	CatBoost	6,436	3	0.293 ± 0.059	0.345 ± 0.049	3.18
Forward in time	Ridge	6,436	3	0.227 ± 0.075	0.286 ± 0.087	3.34
Forward in time	Mean	6,436	3	-0.155 ± 0.027	-0.155 ± 0.027	4.57

Table 13: Forward-in-time windows (pooled CatBoost, R^2): latest training date, earliest test date, record counts and tier results.

Win.	Train until	Test from	n_{train}	n_{test}	$\mathcal{M}_1$	$\mathcal{M}_2$
1	2017-12-26	2018-05-02	18,529	6,502	0.284	0.335
2	2018-07-20	2018-12-12	24,074	6,322	0.356	0.398
3	2019-01-27	2019-07-19	29,429	6,484	0.240	0.302

$\mathcal{M}_1 \rightarrow \mathcal{M}_2$ gain in that experiment. The property values may still carry information that this design cannot separate from identity, but transferable physical information is not demonstrated; a leave-material-out test would be needed. The result is consistent with prior work showing that the representation of material and process information strongly affects PCE prediction[11], and it shows that the benefit persists when entire publications are held out.

The same five-holdout protocol was used for the earlier 44-predictor literature representation (Supplementary Information, Tables S2 and S5). Added to $\mathcal{M}_2$ it produced no measurable change ($\Delta R^2 = -0.001$ pooled), and that result, together with the redundancy of the extracted numerical parameters with PDP columns (Section 2.2.2), motivated the more targeted recipe-flag representation. It does not bear on the size of the recipe-flag effect, which is evaluated only in Sections 3.1 and 3.2.

Table 14: Paired difference between $\mathcal{M}_1$ and $\mathcal{M}_2$ (matched CatBoost, five DOI-grouped splits). ΔR^2 and ΔRMSE are means over the five splits; "better" is the number of splits in which R^2 improved; "CI≠ 0" is the number of splits in which the 95% DOI-cluster bootstrap interval of the paired ΔR^2 excludes zero. Test sets overlap across splits, so no pooled significance test is reported.

	$\mathcal{M}_1 \rightarrow \mathcal{M}_2$			
Subset	ΔR^2	ΔRMSE	better	CI≠ 0
Pooled	+0.040	−0.125	5/5	5/5
p-i-n	+0.047	−0.130	5/5	2/5
n-i-p	+0.029	−0.093	5/5	3/5

3.7. Interpretation of model predictions

SHAP values are attributions for the fitted model and do not demonstrate predictive benefit or causal effects. The SHAP analysis was computed for models trained on $\mathcal{M}_2$ plus the earlier 44-predictor literature

representation and not for the recipe-flag model $\mathcal{M}_3$; its figures and rankings are given in the Supplementary Information (Figs. S1 and S2) and are not evidence about $\mathcal{M}_3$. In those models the PDP binary field for perovskite quenching or crystallisation (`Perovskite_deposition_quenching_induced_crystallisation`) was the highest-ranked predictor, and aggregating importance by feature group showed material-property (33–38%) and process (36–42%) variables dominating, followed by composition (13–15%) and device (6–8%) descriptors. Because material properties are fixed functions of material identity, layer-property features such as HTL band gap also act as identifiers of the material used in that layer. We did not compute SHAP attributions for the 36 recipe flags.

3.8. Comparison with previous PDP-based studies

Table 15 summarises what the two most closely related PDP studies did and reported. The numbers are not comparable with ours. Roberts et al. predict short-circuit current density (J_{SC}, with the RMSE in mA cm^{-2}), whereas we predict PCE (errors in percentage points); the two targets and units are different, so their absolute metrics cannot be set against ours. Fukasawa et al. predict PCE, but their record set (36,937 curated records, 248 variables), feature sets and test sets differ from ours, and their conventional-split result and their new-record result are different evaluations that Table 15 keeps apart. As an indirect, like-for-like reference we ran the *settings* associated with these studies inside our framework (Table 9): XGBoost with library defaults (the setting of Roberts-style pipelines) and a random forest (the best model of Fukasawa et al.) on the database-only tier $\mathcal{M}_1$, evaluated on the same DOI-grouped splits. They reach $R^2 = 0.334$ and 0.295, against 0.368 for matched CatBoost on $\mathcal{M}_1$. These are not re-implementations of the original feature engineering (one-hot layered encodings and duplicate averaging for Roberts et al.; Oliynyk composition vectors and zero complement for Fukasawa et al.), and we did not re-run the original pipelines.

Table 15: Related PDP-based studies as described in the cited articles, and this study. Targets and units differ: Roberts et al. predict J_{SC} (RMSE in mA cm^{-2}) and this study predicts PCE (MAE and RMSE in percentage points), so those metrics are not comparable. Fukasawa et al.'s conventional-split and new-record results are separate evaluations on different test sets. The two rows for this study are different evaluations (a single reserved-publication partition with all three tiers, and a five-split repeated holdout with $\mathcal{M}_2$) and should not be mixed.

Study	Target	Partitioning	Model and reported performance
Roberts et al.[9]	J_{SC}	80/10/10 train/validation/test; duplicates averaged	XGBoost: $R^2 = 0.35$, RMSE = 3.58 mA cm^{-2}
Fukasawa et al.[11]	PCE	8:2 train–test split of 36,937 curated records (no grouping by publication described), 25% of training data for validation; separate evaluation on 294 newly registered records	Random forest (best model): conventional split $R^2 = 0.7177$; new records $R^2 = 0.45$, MAE = 3.20% (mean model on the new records: $R^2 = -0.33$, MAE = 5.40%)
This study, reserved publications	PCE	DOI-grouped holdout (2,619 reserved publications), DOI-disjoint validation	Matched CatBoost, all 14,125 records, $\mathcal{M}_3$: $R^2 = 0.406$, MAE = 2.95, RMSE = 3.79 (LLM-covered, 5,460 records: $\mathcal{M}_3$ $R^2 = 0.366$)
This study, five-split holdout	PCE	DOI-grouped repeated holdout (five splits), DOI-disjoint validation	Matched CatBoost, pooled, $\mathcal{M}_2$: $R^2 = 0.408 \pm 0.032$, MAE = 2.95 ± 0.08, RMSE = 3.78 ± 0.09

3.9. Limitations

Evaluation. Results come from a single database with heterogeneous reporting, and 69% of the PCE variance lies between publications, so held-out R^2 near 0.4 means that most between-paper variation is

not explained by the available descriptors. The five repeated DOI-grouped test sets overlap, so the reported spreads describe split-to-split variation and not an independent repeated cross-validation, and the publication-cluster bootstrap intervals address within-test-publication dependence but not variability due to training-set choice or model selection. DOI grouping does not separate publications from the same research group; lead-author surname grouping is a coarse proxy, and the forward-in-time windows rely on a database date field and are affected by distribution drift. Neither is an external validation on independently collected devices, and the models were not assessed for tandem or non-standard devices or for ranking high-efficiency devices, which are systematically under-predicted. Devices with PCE below 1% were excluded, which selects for functioning devices. Some publications report the same device configuration (1,277 records in 162 groups of identical predictor vectors of the earlier representation span more than one DOI), and such cross-publication near-duplicates can fall on both sides of a split. The 232 records without a DOI were kept together, which is conservative but uninformative about their true publications. Reduced-illumination records remain in the data because illumination is not a predictor; test-set exclusion changes scores by about 0.005.

Recipe flags. The flags are available for 36.6% of records, are paper-level, and cannot separate devices within a publication. They are broad lexical matches over LLM-extracted records (for example the ammonium-salt flag also fires on methylammonium), they have not been checked against a manually annotated gold set (a worksheet exists but has not been reviewed), the extraction model checkpoint and quantisation are not recorded, and no non-LLM text-mining baseline was built, so we do not claim that they contain information absent from the PDP. The evidence for the $\mathcal{M}_2 \to \mathcal{M}_3$ step rests on one reserved partition of about 1,000 covered publications (resolution roughly ± 0.01 R^2); the flag block was chosen on development comparisons, the reserved publications had been examined in earlier experiments, all pre-specified intervals include zero, and the size of the effect depends on the CatBoost configuration. The $\mathcal{M}_3$ models also carry a first-feature-use penalty on the flag columns. The five-holdout experiments, the reference models, the model-development ensemble, the stricter protocols and the SHAP analysis do not include $\mathcal{M}_3$.

Material properties. The descriptors are material-level reference values that identify materials as much as they describe them, the material identity control attributes essentially the whole $\mathcal{M}_1 \to \mathcal{M}_2$ gain to identity, the lookup takes the first non-missing value per property and layer family so that one layer's descriptor vector can mix materials, per-value sources are not auditable from the supplied files, and the atomic-radius columns are covalent-radius composition summaries and not ionic radii. The reference Gaussian process was fitted to a subsample and the MLP was not extensively tuned, so the reference-model table does not rank model families.

Corpus and reproducibility. The route by which each historical PDF was obtained was not logged. Predictions saturate near 21% PCE, can be negative, and under-predict high-efficiency devices.

4. Conclusion

We built a PCE prediction pipeline for the Perovskite Database Project and evaluated it on unseen publications with DOI-grouped, validation-based holdouts. Three results stand out. First, the splitting protocol strongly affects the measured score: record-level splitting gives a CatBoost R^2 of 0.59–0.62 against 0.37–0.41 for DOI-grouped splitting of the same data and model family, whereas a ridge model shows a gap of 0.06. Together with the finding that 69% of PCE variance lies between publications, and the longer training that record-level fits require, this is consistent with publication-specific memorisation, and we recommend reporting DOI-grouped (or stricter) estimates alongside record-level ones for literature-compiled PSC data. Forward-in-time evaluation lowers R^2 further to 0.345 (MAE 3.18), and the results do not establish performance on independently collected data.

Second, curated material-property descriptors improve held-out prediction (R^2 0.371 to 0.403 on all 14,125 reserved records, and 0.318 to 0.359 on the covered records), but a material-identity control reproduces the gain (0.406 versus 0.407), so the improvement is attributable to knowing which materials are present and not to demonstrated independent physical information.

Third, adding 36 binary recipe flags built from LLM-extracted full-text information improved the observed held-out predictions under matched settings: R^2 rose from 0.359 to 0.366 (MAE 2.911 to 2.892, RMSE

3.747 to 3.726) on the 5,460 reserved records whose publications have extraction data, and from 0.403 to 0.406 (MAE 2.955 to 2.949, RMSE 3.794 to 3.787) on all 14,125 reserved records. The gain is larger where the flags exist, exceeds five permuted-flag controls and a coverage-indicator control, and persists in point estimate under publication weighting and without the first-feature-use penalty. It is small, the pre-specified intervals ($\Delta R^2 = +0.0071$, -0.0007 to $+0.0157$ covered; $+0.0025$, -0.0011 to $+0.0062$ all records) include zero, it depends on the estimator configuration, and it was measured on one reserved partition with limited coverage; its magnitude and generalizability therefore remain to be established, for example by extracting the roughly 4,500 publications that currently lack extractions and by validating the flags against a manually annotated set. The best validation-tuned CatBoost–XGBoost ensemble on $\mathcal{M}_2$ reached a pooled R^2 of 0.416 (MAE 2.92) in the five-holdout experiment, with substantial architecture dependence (n-i-p 0.41, p-i-n 0.30) and a split-to-split standard deviation of about 0.03. The code, configurations, tests and per-split predictions allow the reported numbers to be regenerated from the supplied table.

Data and code availability

The accompanying repository, supplied for peer review and to be archived on Zenodo with DOIs given in this statement at acceptance, contains the code, configuration files, tests, run manifests, per-split and reserved-split predictions and metrics needed to regenerate every table and figure. The modelling inputs it supplies are: the processed modelling table (43,398 PDP records with device-level descriptors, material-property columns and the 32 base columns of the earlier literature representation; part of the review package and its SHA-256 is recorded in every run manifest, but excluded from the Git history because of its size); the per-publication derived literature artifact keyed by normalised DOI (the tokens, aggregate numeric values, layer records and the 36 recipe flags built from the cached LLM extractions); the PDP reference metadata used for lead-author and date grouping; and the per-predictor audit of the three tiers. The curated material-property catalogue is not supplied as a separate file; its values enter the processed table as columns. The PDP is openly available under a CC BY licence[8]. The publisher PDFs, their Markdown conversions and the raw LLM output files are not redistributed because they derive from copyrighted full texts; the DOI list is derivable from the PDP. The extraction prompt and settings are given in the Supplementary Information.

Declaration of AI use

A locally served large language model (Qwen) was used as part of the scientific method (literature feature extraction, Section 2.2) and is described above. During the preparation of this work the author(s) used Claude in order to audit the original code, format contents and correct language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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